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Roll No. ....

Level 3, 2nd Sem / DVOC (Ref. & Air Cond., Medical Imaging Tech., Auto Servicing, ITM, PT, SD, AMT, FP, EMS)

Subject : Basic Electronics

Time : 2 Hrs. M.M. : 50

SECTION-A

Note: Multiple Choice questions. All questions are compulsory. (5x1=5)

Q.1 What is the charge of an electron?

a) Positive b) Negative

c) neutral d) none of the above

Q.2 Which component is used for rectification?

a) resistor b) diode

c) inductor d) capacitor

Q.3 What is the primary function of a capacitor?

a) store charge b) generate current

c) regulate voltage d) convert ac to dc

Q.4 In a PN junction diode, the depletion layer is formed due to :

a) movement of electrons

b) movement of holes

c) combination of electrons and holes

d) external voltage application

Q.5 Which of the following is a unidirectional device?

a) resistor b) inductor

c) diode d) transformer

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

Q.6 Define the Ohm's law.

Q.7 What is the full form of CRO?

Q.8 What is a semiconductor?

Q.9 Name two types of transistors.

Q.10 What is the function of a zener diode?

SECTION-C

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

Q.11 Differentiate between AC and DC current.

Q.12 What are the applications of a PN junction diode?

Q.13 Define intrinsic and extrinsic semiconductors.

Q.14 Explain the working of a transistor in CE mode.

Q.15 What are the differences between CB, CE, and CC transistor configurations?

Q.16 What is the significance of the depletion layer in a diode?

Q.17 Explain the working of a simple rectifier circuit.

Q.18 What is the difference between a conductor and a semiconductor?

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Explain in detail the working of a Bipolar Junction Transistor (BJT) and its characteristics.

Q.20 Describe the operation of a full-wave bridge rectifier with a neat circuit diagram.